

MONTEGO BAY SANGSTER, Jamaica

WMO: 783880

Lat: 18.504N

Lon: 77.913W

Elev: 1

StdP: 101.31

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.9	22.7	15.9	11.3	24.1	17.1	12.2	24.4	11.3	26.6	10.3	26.6	2.0	140	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	33.0	27.2	32.2	27.0	32.0	26.9	28.7	31.5	28.1	30.9	27.8	30.6	7.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	24.1	30.9	27.2	23.0	30.2	27.0	22.7	30.1	93.1	31.3	90.1	31.3	88.6	30.8	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.1	9.2	8.4	DB	20.7	34.5	1.2	0.5	19.9	34.9	19.2	35.2	18.5	35.5	17.6	35.9	
			WB	17.4	28.8	1.2	0.8	16.6	29.4	15.9	29.9	15.2	30.4	14.3	30.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	25.8	25.9	26.2	27.0	27.7	28.6	28.9	29.0	28.7	28.1	27.4	26.5
	DBStd	1.46	0.98	0.86	1.00	0.79	0.89	0.90	0.78	0.84	0.88	0.93	0.97	1.02
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6387	490	445	502	511	549	558	586	588	561	562	523	512
	CDD18.3	3345	232	212	243	261	291	308	328	329	311	303	273	254
	CDH23.3	35154	1805	1682	2080	2602	3139	3705	4030	4004	3648	3374	2812	2275
Wind	WSAvg	3.4	3.9	3.8	3.7	3.5	3.2	3.4	3.6	3.2	2.7	2.8	3.5	3.8
	PrecAvg	1135	60	46	57	66	131	106	67	101	148	171	116	106
	PrecMax	1653	234	201	166	326	447	330	297	305	423	674	358	380
	PrecMin	619	11	0	0	0	0	6	15	22	30	22	16	16
	PrecStd	287	51	45	49	71	94	74	56	63	78	124	78	83
	DB	30.1	30.2	31.0	31.2	32.1	33.3	33.4	33.7	33.2	32.8	32.0	31.1	
	MCWB	24.4	24.3	25.1	25.5	26.4	26.6	27.5	27.6	27.8	27.3	27.2	26.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	29.2	29.6	30.0	30.3	31.2	32.3	32.7	32.8	32.2	31.8	31.0	30.1	
	2% MCWB	24.5	24.2	24.8	25.4	26.4	26.7	27.0	27.5	27.2	27.0	26.8	25.5	
	5% DB	28.9	29.1	29.2	30.0	30.9	32.0	32.1	32.1	31.9	31.1	30.2	29.3	
	5% MCWB	24.5	24.2	24.5	25.3	26.3	26.7	26.9	27.1	27.1	26.8	26.2	25.1	
	10% DB	28.2	28.5	28.9	29.4	30.2	31.2	31.6	31.7	31.1	30.8	29.9	29.0	
	10% MCWB	24.1	23.9	24.4	24.9	25.9	26.5	26.8	27.0	26.9	26.6	26.0	25.0	
	WB	26.6	26.1	26.7	27.5	28.5	28.5	29.1	29.2	29.1	28.8	28.4	27.8	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	28.1	28.4	28.4	29.4	30.0	31.4	32.3	32.4	32.2	31.5	30.3	29.4	
	2% WB	25.9	25.5	26.0	26.7	27.6	27.9	28.2	28.4	28.2	27.9	27.6	26.5	
	2% MCDB	27.8	28.1	28.4	29.2	29.8	30.9	31.3	31.5	31.1	30.3	29.7	28.5	
	5% WB	25.2	25.0	25.4	26.1	27.0	27.4	27.7	27.9	27.8	27.3	27.0	26.0	
	5% MCDB	27.7	27.9	28.2	28.9	29.6	30.7	30.9	30.9	30.7	30.0	29.3	28.4	
	10% WB	24.7	24.5	24.9	25.6	26.5	26.9	27.2	27.3	27.3	26.9	26.5	25.5	
	10% MCDB	27.4	27.6	27.9	28.6	29.3	30.4	30.6	30.7	30.4	29.7	28.9	28.1	
Mean Daily Temperature Range	MDBR	5.1	5.6	5.5	5.4	5.5	6.0	6.0	6.1	5.8	5.4	5.0	5.0	
	5% DB	5.8	6.2	6.0	5.9	6.0	6.5	6.5	6.5	6.3	5.9	5.6	5.6	
	MCWBR	3.7	4.0	3.8	3.7	3.9	4.1	4.1	4.0	3.8	3.7	3.3	3.5	
	5% WB	5.2	5.8	5.7	5.5	5.6	6.0	6.1	6.1	5.8	5.4	5.3	5.2	
	MCWBR	3.7	4.1	3.9	3.7	4.0	4.2	4.3	4.1	4.0	3.7	3.5	3.5	
Clear-Sky Solar Irradiance	taub	0.380	0.382	0.388	0.420	0.447	0.491	0.500	0.462	0.423	0.394	0.387	0.378	
	taud	2.533	2.511	2.491	2.403	2.364	2.228	2.186	2.335	2.453	2.554	2.540	2.560	
	Ebn at Noon	904	920	921	888	853	809	803	840	876	896	889	894	
	Edh at Noon	99	105	111	122	125	142	149	129	114	100	97	94	
All-Sky Solar Radiation	RadAvg	4.42	5.20	5.87	6.28	5.95	5.89	6.17	5.98	5.58	4.86	4.47	4.21	
	RadStd	0.26	0.22	0.21	0.24	0.34	0.31	0.35	0.20	0.41	0.39	0.31	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page